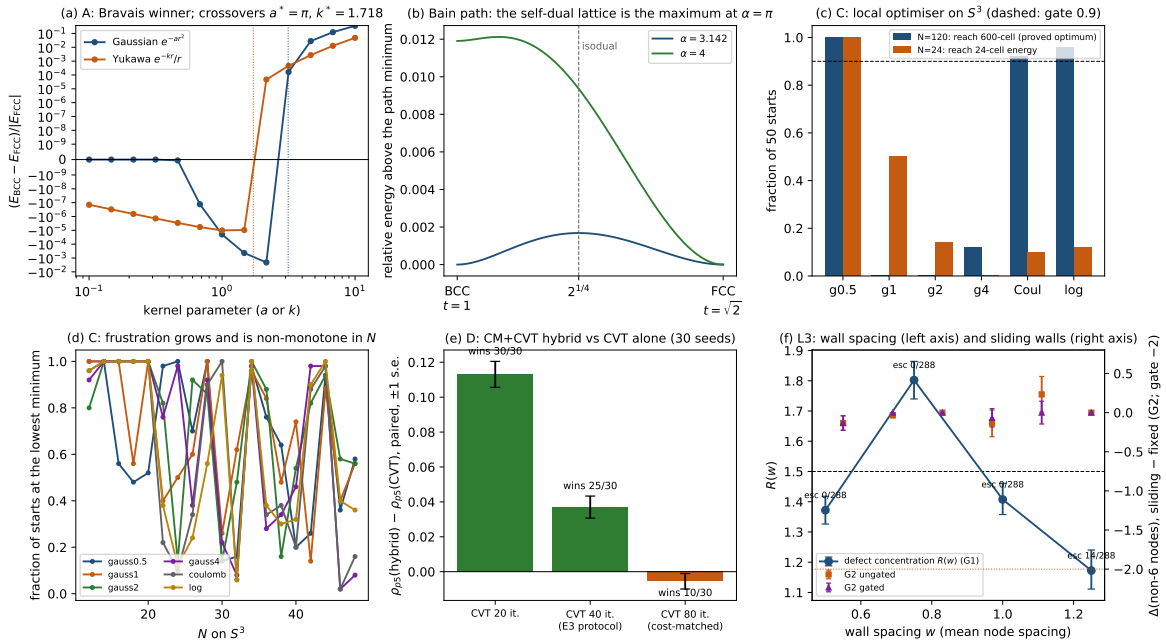


(a) Which Bravais lattice wins in 3D for Gaussian and Yukawa kernels at unit density; the crossovers are $a^* = \pi$ (exact, Poisson duality) and $k^* = 1.718$. (b) Gaussian energy along the Bain path between BCC and FCC: at the self-dual point $\alpha = \pi$ the self-dual (isodual) lattice is the *maximum*. (c) Fraction of 50 random starts that a local optimiser brings to the proved optimum on S^3 (600-cell, $N = 120$) and to the 24-cell energy ($N = 24$). (d) Fraction of starts reaching the lowest minimum found, $N \le 48$ on S^3. (e) Hybrid CM+CVT mesh against CVT alone, paired over 30 seeds. (f) Wall-spacing scan and sliding walls in the Fenris container study.

Gated exciton fluids V:

three dimensions — what the T-duality, holography and projection analogies do and do not give, measured

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October 2026 (version 2: adds H1′; version 3: adds H1″ and H1‴, certified bounds; version 4: adds H1d, certified upper bounds)

Abstract

Papers I–IV [1, 2, 3, 4] proved that the gated dipole kernel is completely monotone in r^2 and tested, negatively, three heuristics for the two-dimensional node problem. This paper turns to three dimensions, where the owner asked whether ideas borrowed from string theory — lifting to 8 or 24 dimensions, holography, projection, T-duality — could "solve" the problem. We first state the known fact that reframes the question: in $\mathbb{R}^3$ no configuration is universally optimal, because the Poisson dual of FCC is BCC and the two exchange roles at the self-dual Gaussian parameter $\alpha = \pi$; the open problems are the Gaussian-core conjecture (FCC/BCC among *all* configurations), crystallisation, and mixed kernels. We then report pre-registered experiments. Round 1: among 3D Bravais lattices the minimiser is always FCC or BCC (29/29 kernel points; crossovers $a^* = \pi$ to 3×10^{-14} and $k^* = 1.7179$ for Yukawa); random-restart searches over non-Bravais cells with up to 16 points find nothing below min(FCC, BCC, HCP, A15) in 9 860 restarts; on S^3, where the 600-cell is a *proved* optimum, a local optimiser reaches it in 46–50 of 50 starts for Coulomb, logarithmic and soft Gaussian kernels but in 0 and 6 of 50 for stiffer Gaussians; a completely-monotone pre-relaxation improves a CVT mesh only when the CVT is given less work; sliding container walls reduce the defect count in 0 of 6 shapes. A direct computation along the Bain path shows that at $\alpha = \pi$ the self-dual (isodual) 3D lattice is the energy *maximum* between the

two degenerate minima: the T-duality analogy explains why no universal optimum exists in 3D rather than pointing to one. The mode-sum ("Kaluza–Klein tower") evaluator of the gated kernel is 4–7 times cheaper than a tail-corrected image sum. Round 2 is negative throughout: an α-homotopy from a soft kernel reaches the proved 600-cell in 0, 0 and 6 of 50 seeds at $a = 1, 2, 4$, while plain restarts at equal compute reach it in 50/50 at $a = 4$ (it does help for the 24-cell), so by the registered rule our 3D searches are declared unreliable; Gaussian mixtures straddling the self-dual point produce nothing below FCC/BCC/HCP in 5 100 restarts; a search for formally self-dual configurations accepts nothing, but its objective cannot tell the known one (diamond) from configurations 10^{-3} away, so the negative is uninformative; and the two-point linear-programming code failed its own one-dimensional sanity gate (under an admissibility tolerance fixed in code, not registered) and was therefore not run in $d = 3$. *Version 2* adds H1′: the $\alpha = \pi$ question of H1 turned out to be answered in the literature (two-point bound 0.2232 against 0.2315 in $d = 3$); after an attempt at rigorous certificates was abandoned in development, a re-registered *numerical, not certified* run reproduces the published $d = 1, 2, 3$ values within the registered windows and gives gaps of 0.48%, 6.7% and 21.6% below the best lattice in $d = 3$ at $\alpha = 2, 4, 8$: numerically, at polynomial degree ≤ 40 and with the best lattice as reference, two-point bounds do not settle the 3D question at any α tested. *Version 3* adds interval-certified bounds (256-bit interior point, exact cutting planes, arb certificate): at $K = 40$ (polynomial degree 39) the $d = 3$, $\alpha = \pi$ gate failed by 3×10^{-8} (H1″); at $K = 50$ (degree 49) it passed (H1‴, certified 0.2232178194 against the published 0.22321782), and the certified lower bounds at $\alpha = 2, 4, 8$ lie 0.48%, 6.7% and 21.6% below the best lattice. The bound values are rigorous; that two-point bounds cannot do better remained numerical in version 3 (version 4 certifies it, below). A review found a defect in our certificate code (sub-ulp negative intervals of $\hat{f}$ not checked); an independent re-check confirms every reported bound, one of them ($K = 60$, not a gate) only after adding a small lift. *Version 4* adds the missing direction: rigorous *upper* bounds on the two-point bound, by an energy adaptation of Li's discrete-reduction dual. They lie below $\min(E_{\mathrm{FCC}}, E_{\mathrm{BCC}})$ by 0.414%, 5.82% and 18.3% at $\alpha = 2, 4, 8$, so the two-point bound (in the form of Cohn–de Courcy-Ireland's Prop. 2.2, whose full class is reached by a lemma added after registration) provably cannot prove FCC or BCC optimal there, and at $\alpha = \pi$ the upper bound is below the known coexistence record; with version 3 the two-point bound is bracketed, e.g. in $[0.2232178, 0.2242867]$ at $\alpha = \pi$. Everything is small-scale, classical and single-engine; nothing uses the unrefereed planar lattice theorem; written with an AI system and not refereed.

1 The question, restated

For a unit-density lattice $L \subset \mathbb{R}^3$ and the Gaussian kernel $e^{-\alpha r^2}$, Poisson summation gives $\theta_L(\alpha) = \sum_{v \in L \setminus 0} e^{-\alpha |v|^2}$ and $1 + \theta_L(\alpha) = (\pi/\alpha)^{3/2} (1 + \theta_{L^*}(\pi^2/\alpha))$. The dual of FCC is BCC. Hence $\theta_{\mathrm{FCC}}(\pi) = \theta_{\mathrm{BCC}}(\pi)$ exactly, BCC is lower for $\alpha < \pi$ and FCC for $\alpha > \pi$ (Fig. (a); our lattice search finds the crossover at $a^* = 3.14159265358987$). By Bernstein's theorem every completely monotone kernel is a positive mixture of Gaussians, so no single configuration can minimise all of them: **universal optimality is false in $\mathbb{R}^3$ for every configuration**. This is known [6, 7]; we state it because it voids the phrase "solve the Cohn–Kumar conjecture in 3D". What is open in three dimensions is

Q1 the Gaussian-core conjecture: BCC ($\alpha < \pi$) and FCC ($\alpha > \pi$) minimise among *all* periodic configurations ("no proof is known" [7]); at $\alpha = \pi$ itself the lowest known energy ("putative ground state") is a Maxwell coexistence of the two at densities $0.99899854 \leq \rho \leq 1.00100312$, about 4×10^{-6} (relative) below either pure lattice [7];

Q2 crystallisation (periodicity of minimisers) for soft repulsions;

Q3 kernels whose Bernstein measure charges both sides of π, for which nothing excludes a non-Bravais minimiser.

"Borrowing from 8D/24D" can therefore only mean borrowing *proof mechanisms* — self-duality,

Fourier interpolation, linear-programming bounds — not answers. The experiments below give each analogy a falsifiable form.

What was pre-registered. Round 1 (`exploration/exciton/three_d/PREREG.md`, commit `357ee6d`): A (Bravais), B (non-Bravais cells), C (S^3), D (hybrid mesh); L3 (`fenris-gated/L3_PREREG.md`, see the caveat in §2.5); L1 (`exploration/exciton/kk/PREREG.md`). Round 2 (`PREREG_round2.md`, commit `51a85b1`): F1, T2, T1, H1. Version 2: H1′ (`three_d/H1p/PREREG.md`, `d668109`; amendments A1 `83a3500`, A2 `968dd32`, both before any gate run, but A2's acceptance tolerance and windows were set after H1's outcomes were known). Version 3: H1″ (`three_d/H1pp/PREREG.md`, `b1d2de5`) and H1‴ (`three_d/H1ppp/PREREG.md`, `1e50f4b`; evaluator replaced during the runs, `162dfc3`, see `H1ppp/DEVIATIONS.md`). Version 4: H1d (`three_d/H1d/PREREG.md`, `4e596f0`; evaluator tightened during the runs, `1ee1714`, see `H1d/DEVIATIONS.md`). All gates, restart counts and thresholds were fixed before the runs, except that H1d's evaluator was tightened during the runs (a 10^{-12} margin and stricter voiding); deviations are dated in each `RESULTS.md` and summarised in §10. An independent adversarial review preceded this write-up; its corrections are applied and listed.

2 Round 1

2.1 A: Bravais lattices

Free 9-parameter unit-volume cell with periodic Minkowski re-reduction, 300 random starts per kernel point, Ewald energies with tails $\leq 8 \times 10^{-16}|E|$; kernels Gaussian ($a \in$ 13 log-spaced points in $[0.1, 10]$), Yukawa (k, same grid), Riesz ($s \in \{4, 6, 12\}$). Gate A1 (global Bravais minimiser = lower of FCC/BCC at $\geq 95\%$ of points): **met, 29/29**. Five Gaussian points ($a \leq 0.46$) are degenerate in double precision (FCC, BCC, A15, C15 and diamond agree to 10^{-9}), so the pass is vacuous there; 24/29 non-degenerate points still exceed 95%. Crossovers: Gaussian $a^* = \pi$ (exact by duality), Yukawa $k^* = 1.7179$ with a relative gap of only 10^{-7}–10^{-5} around it, Riesz FCC throughout $s \in [3.05, 30]$. Verification: analytic gradients against finite differences (2×10^{-9}), Gaussian energies against closed-form theta functions ($\leq 3 \times 10^{-13}$), Yukawa/Riesz against brute-force sums with continuum tails and an independent Ewald code (10^{-15}).

2.2 B: non-Bravais cells

$N \in \{2, 4, 8, 12, 16\}$ points per free cell; 100 restarts for $N \leq 4$, 60 for $N = 8$, 40 for $N = 12, 16$ (reduced from the registered 100 for time; disclosed). References FCC, BCC, HCP, diamond, A15 (all representable at the relevant N), C15 ($N = 12$ only). Gate B1 (restarts reach the best reference in $\geq 50\%$ of cases for $N \leq 4$, $\geq 10\%$ for $N = 16$): **not met for $N \leq 4$ under the strictest reading** (49/58 gated cells pass; the 9 failures sit at 39–49%, all near the FCC/BCC crossovers where the two basins compete; the median hit fraction, 0.76 at $N = 2$ and 0.66 at $N = 4$, would pass a pooled reading); met for $N = 16$ (29/29, minimum 25%). Gate B2 (discovery): **nothing** below min(FCC, BCC, HCP, A15) by more than 10^{-8} in 9 860 restarts over 145 cells. A relaxed HCP/DHCP stacking scan is never below FCC/BCC by more than 10^{-9}, consistent with the statement that FCC is the best Barlow packing for the low-density Gaussian core [8]. The first A runs were invalid (a determinant-sign bug, two overlapping runs, a hang on skewed cells); their records are kept in `run0_buggy/` and the reported numbers are the fourth run.

2.3 C: a proved control on S^3

N points on the unit S^3, chordal kernels e^{-ad^2} ($a \in \{0.5, 1, 2, 4\}$), $1/d$ and $\log(1/d)$; L-BFGS-B from 50 random starts per kernel. The 600-cell is universally optimal for $N = 120$ [6], a *proved* target. Gate C1 ($\geq 45/50$ starts within 10^{-8} of E_{600} for every kernel): **not met**. Success 50/50 ($a = 0.5$), 46/50 (Coulomb), 48/50 (log), 6/50 ($a = 4$, 44 starts end 10^{-3} above; round 2 indicates a low draw: restarts of the same optimiser succeed in 1 916 of 5 790 runs, fresh starts in 38/140, Sec. 5.1), 0/50 ($a = 2$, all converge to a family of local minima 7×10^{-6} above). For $a = 1$ all 50 starts hit the registered 8 000-iteration cap 2–3×10^{-8} above E_{600}; the review asked for a rerun with the cap raised to 150 000: all converge (16 640–66 673 iterations) at 1.5–2.1×10^{-8} above, and two polished configurations have minimum chord 0.50 against 0.618 for the 600-cell with a distance-spectrum deviation of 0.22 — *distinct local minima*, not the 600-cell with residual error, in a landscape so flat that the best local minima lie within 2×10^{-8} of the proved optimum. No start went below E_{600}. For $N = 24$ the 24-cell energy is reached in 50, 25, 7, 0, 5, 6 of 50 starts (same kernel order); the 24-cell is known *not* to be universally optimal and to have a small basin of attraction [9], and our Coulomb and log runs reproduce the latter qualitatively with a different potential. Frustration scan: a second distinct minimum appears at $N = 12$ (stiff Gaussians, Coulomb, log), 16 ($a = 0.5$), 18 ($a = 1$); non-uniqueness is common from $N = 22$ and non-monotone in N (Fig. (d)). Seeds for ($N = 120$, kernel 0) and ($N = 20$, kernel 1) coincide (a formula artefact; correlated, not independent, streams).

Reading. Where the optimum is proved and the kernel is long-ranged or soft, a plain local optimiser finds it; where the kernel is stiffer it is trapped. The obstacle of the three-dimensional problem in our hands is the landscape, not the dimension. Any method that claims to help in 3D must first pass this control; round 2 (F1) is built on it.

2.4 D: a CM pre-relaxation as a mesh accelerator

Continuation of E3 of paper IV with the same $N = 216$ periodic cube and the registered 30 seeds: CM relaxation followed by 20 Lloyd iterations against CVT alone. "Equal total iterations" had not been defined in the pre-registration; the run's author fixed CVT-40 as primary before running and added CVT-20 and CVT-80 as sensitivity arms. Hybrid median 5th-percentile radius ratio 0.7057: against CVT-40 (the E3 protocol) $+0.0370 \pm 0.0063$, wins 25/30; against CVT-20 $+0.113 \pm 0.007$, 30/30; against CVT-80, the *cost-matched* arm (CM stage ≈ 60 s, hybrid ≈ 82 s, CVT-80 ≈ 88 s), -0.0055 ± 0.0045, wins 10/30. **Verdict: met against CVT-20/40, not met at matched cost**; the hybrid is a convergence accelerator at best. It does not improve the worst element (worst ρ 0.014 vs 0.029, worst minimum dihedral 0.73° vs 1.44°). Seed 0 was a development seed inside the registered set; without it, $+0.0371 \pm 0.0065$, 24/29.

2.5 L3: wall spacing and sliding walls in the Fenris study

A fresh clone of `rusty-Fenris-CMRR` (commit `6d0df125`, never pushed) with the kernel patch of paper III. G1 (wall-spacing scan $w \in \{0.5, 0.75, 1, 1.25\}$, random-sequential-addition starts, 6 shapes $\times$ 2 kernels $\times$ 24 starts): 0/288 escapes at $w \leq 1$, 14/288 at $w = 1.25$. **The registered $w = 1$ wall confines with RSA starts**, so the "did not confine" finding of paper III came from the uniform starts, not from the spacing; the saddle mechanism was not tested. Defect concentration $R(w) = 1.37, 1.80, 1.41, 1.17$; the registered threshold 1.5 is met only at $w = 0.75$ and that value is confounded by the triangulation-site rule (all wall nodes used as sites at $w \neq 0.5$). G2 (engine-level sliding walls, $w = 0.5$, 30 paired starts per shape): **0 of 6 shapes** reach a decrease of 2 defects for either kernel (square -0.13 ± 0.09, rect -0.14 ± 0.17, L-shape $+0.23 \pm 0.23$, others ≈ 0); the wall moves by at most 0.05; the wall-coordinate L-BFGS hit its 40-iteration cap in 25–30 of 30 starts for two shapes, so this is weak evidence for those shapes.

Compared with E2 of paper IV (joint minimisation from the start) the square has the same sign and a 15 times smaller magnitude. *Caveat:* `L3_PREREG.md` was untracked when the runs were made; its registration time rests on the author's attestation, not on git.

2.6 L1: the Kaluza–Klein tower as an evaluator

The gated kernel of paper I is $U(r) = \frac{4}{\ell}\sum_{m\geq 1}(\varphi_m(z_e) - \varphi_m(z_h))^2 K_0(m\pi r/\ell)$, a mode sum with exponentially decaying terms, or an image sum with algebraic tails. On four parameter sets and 10^4 random $r \in [0.3d, 6\ell]$ the adaptive mode sum (binned in r) costs 0.028–0.036 s against 0.148–0.191 s for the image sum with an Euler–Maclaurin tail ($N = 60$), i.e. 4.1–6.7× ($\geq 5\times$ on 2 of 4 sets, the registered target); the image sum is accurate to 10^{-12} only for $r \lesssim 1.1$–$1.9\,\ell$ because of cancellation. Gate G1 met with a disclosed caveat (mpmath reference at 10 points per set, float modal sum as reference elsewhere, 7.8×10^{-16} agreement), G2 partial; the Rust crate was therefore not built. The "KK tower" is real and useful, and it is exactly Prop. 2 of the mathematical note.

3 The T-duality analogy, computed

In the three cases where universal optimality is known or conjectured (A_2 up to similarity, E_8, Λ_{24}) the minimiser is a fixed point of Poisson duality. Along the body-centred-tetragonal (Bain) path bct(t), $c/a = t$, unit density, $t = 1$ is BCC, $t = \sqrt{2}$ is FCC, and the dual of bct(t) is similar to bct($\sqrt{2}/t$); the fixed point $t = 2^{1/4}$ is isodual. Direct summation (tails $< 10^{-30}$; `AB/bain_selfdual.py`):

α	E(BCC)	E(FCC)	E(isodual)	extremum on the path
π	0.231536266184	0.231536266184	0.231925337063	**maximum at** $t = 2^{1/4}$
2.5	0.526007	0.527283	0.527031	minimum at BCC
4	0.078896	0.077968	0.078697	minimum at FCC

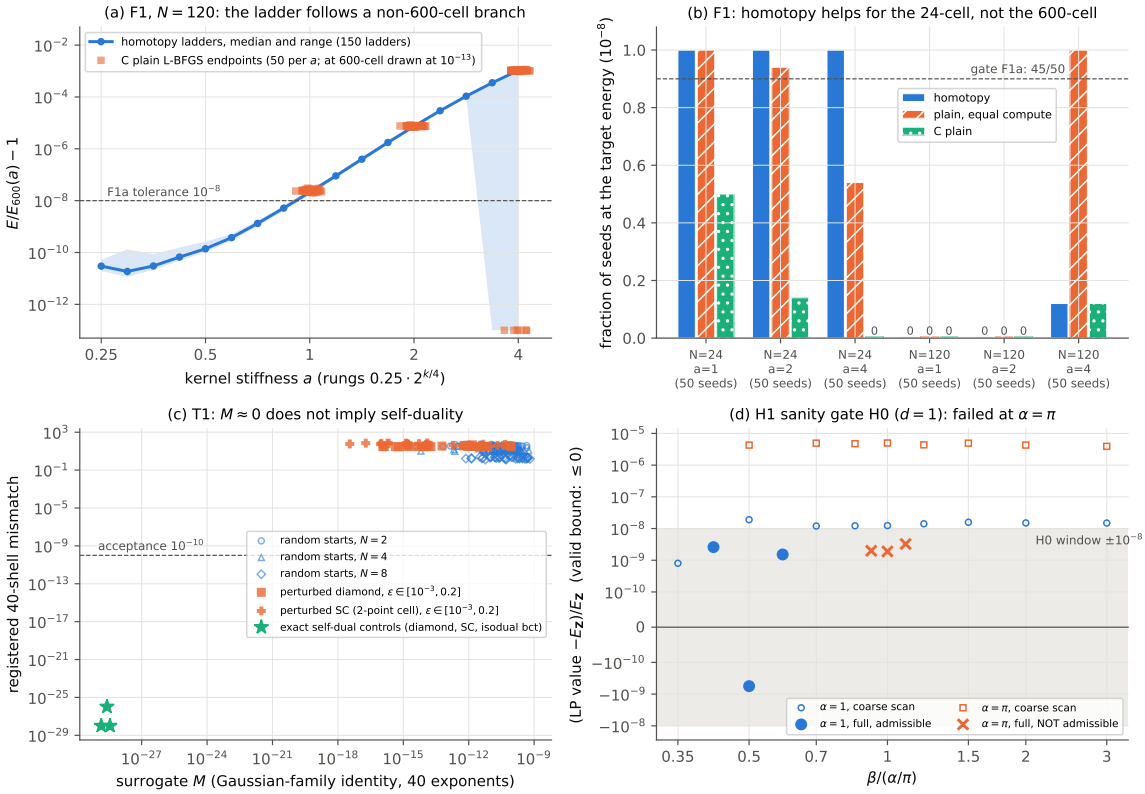
The symmetry $t \leftrightarrow \sqrt{2}/t$ holds to 10^{-16}. **At the self-dual point the self-dual lattice is the worst point of the path**: the duality symmetry is broken into two degenerate minima exchanged by it, the opposite of dimensions 8 and 24. T-duality thus explains the absence of a universal optimum in 3D; it designates no candidate. What survives of the analogy is *formal duality* in the sense of [7]: the diamond structure D_3^+ is formally self-dual, and the question whether any formally self-dual configuration beats FCC/BCC at $\alpha = \pi$ is the content of T1 (§5). Averaged holography over the Narain moduli space gives, by the Siegel mean-value theorem, the *average* Gaussian energy over unimodular lattices, $(\pi/\alpha)^{3/2} - 1$, never the minimum; it is not a tool for this problem.

4 The holography analogy, stated as a question

The proof of universal optimality in dimensions 8 and 24 [10] is holographic in a precise sense: a radial Schwartz function is reconstructed from its values and Fourier values (and derivatives) at the radii $\sqrt{2n}$, through modular forms of level one. In $d = 3$ a radial function is an odd function of the line, and the interpolation nodes $\pm\sqrt{n}$ of [12] are the norms of $\mathbb{Z}^3$ — self-dual, but a poor minimiser. *Does a radial Fourier interpolation formula exist in $d = 3$ with the FCC norms as nodes for f and the BCC norms as nodes for $\hat{f}$?* We have not found this question in the literature and leave it to specialists; even if it exists it cannot prove universal optimality (false), but it could certify FCC at a given α through the linear-programming bound. What we can compute is the size of the two-point LP gap in $d = 3$ (H1, §5; numerical values in version 2, §6).

5 Round 2

Round 2 in one paragraph. F1: the α-homotopy reaches the proved S^3 optimum (600-cell, $N = 120$) in **0, 0 and 6** of 50 seeds at $a = 1, 2, 4$, so gate F1a fails at every a and the registered stop rule applies: the structure searches of this programme are declared unreliable. T2: no structure below FCC/BCC/HCP for Gaussian mixtures straddling the self-dual point (0 candidates in 75 cells, 5 100 restarts). T1: no formally self-dual configuration accepted — but the search cannot distinguish the known one (diamond) from configurations 10^{-3} away, so this negative carries no information. H1: the linear-programming code failed its own sanity gate in $d = 1$ at $\alpha = \pi$ (under an admissibility tolerance fixed in code, not registered) and was therefore not run in $d = 3$ (version 2: numerical $d = 3$ values in Sec. 6).



Round 2. (a) F1, $N = 120$: relative energy above the 600-cell along the α-ladders (line: median; band: range, floored at 10^{-13}; the band reaches the floor at $a = 3.36$, where 2 of the 6 ladders that end at the 600-cell are already there; the other 4 get there at $a = 4$) and C's plain endpoints at $a = 1, 2, 4$ (squares; runs that reach the 600-cell are drawn at 10^{-13}). (b) Fraction of seeds reaching the target energy (energy test only; F1a also requires the chord test, which removes 6 homotopy runs at $N = 24$, $a = 1$, and none elsewhere): homotopy, plain L-BFGS with restarts at equal compute, and C's plain runs at the same seeds. (c) T1: surrogate value against the registered shell mismatch for random starts, perturbed positive controls and the exact controls. (d) H1 sanity gate in $d = 1$: LP value minus $E_{\mathbb{Z}}$ (a valid bound is ≤ 0) over the β scan; crosses are full-setting solutions that miss the admissibility tolerance.

5.1 F1: homotopy in α against the proved control

Same $N = 120$ points on S^3 and kernels e^{-ad^2} as in C; from C's random start at the same seed, L-BFGS "to convergence" at $a_0 = 0.25$, then a geometric ladder of ratio $2^{1/4}$ to the target (rungs 8, 12, 16 for $a = 1, 2, 4$), warm-started. The registration says "re-converged at each rung" and sets no iteration cap; we used C's optimiser settings, including its 8 000-iteration cap (an unregistered choice, listed in the deviations), so the F1a verdict is for the homotopy as implemented with

that cap. Gate F1a: $\geq 45/50$ seeds within 10^{-8} (relative) of E_{600} *and* minimum chord within 10^{-6} of $2\sin(\pi/10)$. Gate F1b: plain L-BFGS from the same start with restarts, hard-capped at the homotopy's number of gradient evaluations (6.2–10.1×10^4), must *not* also reach $\geq 45/50$ at $a \in \{2, 4\}$.

a	homotopy at 600-cell	plain, equal compute	C plain (same seeds)	homotopy end, rel. excess	gradient evaluat
1	0/50	0/50	0/50	$(1.7$–$2.4) \times 10^{-8}$	6.2–7.3×10
2	0/50	0/50	0/50	$(7.0$–$8.2) \times 10^{-6}$	7.8–9.5×10
4	6/50	50/50	6/50	$(0.8$–$1.1) \times 10^{-3}$	8.0–10.1×1

F1a fails at every a**.** The rung-by-rung record locates the failure: already at $a_0 = 0.25$ the state where L-BFGS stops is 3×10^{-11} above E_{600} and is not the 600-cell (one ladder re-run with chords recorded, `F1/supp_rung_chords.py`, seed 0: minimum chord 0.237 against 0.618, distance-spectrum deviation 0.38). It is not a local minimum either: the reviewer found 74 negative eigenvalues of its Hessian on $(S^3)^{120}$, and our own check (`F1/supp_hessian.py`, central differences, $h = 10^{-4}$ and 10^{-5}) finds 83, the lowest -1.8×10^{-6}, with tangent gradient 1.2×10^{-5}; a fresh L-BFGS restart from it stops after 17 iterations, having lowered $E \approx 4445$ by 7×10^{-11}. The remaining descent is near the floating-point resolution of E, so the registered "convergence at a_0" is not reachable in double precision with this energy, and L-BFGS stops on its relative-reduction test at a stall point. The excess then grows smoothly along the ladder (2×10^{-8} at $a = 1$, 7×10^{-6} at $a = 2$, 10^{-3} at $a = 4$), of the same order as C's trapped plain runs (2–3×10^{-8}, 7×10^{-6}, 10^{-3}). By contrast, the trapped plain endpoint at $a = 4$ (seed 0) is a strict local minimum: no negative eigenvalue and exactly six zero modes, the rotations of S^3. So the soft kernel does not select the 600-cell: the soft-kernel landscape is so flat that L-BFGS stops away from it, and the ladder carries that state to stiffer kernels, where it lands on a trapped branch. Six to ten of the 17 rungs (mean 7.8) stop at the 8 000-iteration cap, at every rung from $a = 0.5$ to 1 in every ladder. The relative-excess column is over the seeds that miss. At $a = 4$ the homotopy reaches the 600-cell at 6 seeds (9, 10, 17, 20, 46, 49), none of them among C's six; at each, the ladder sits on the trapped branch ($\approx 10^{-4}$ above at $a = 2.83$) and drops to the 600-cell only at the last one or two rungs ($a \geq 3.36$), so these are late escapes, not branch-following. **F1b** (registered meaning: if plain L-BFGS at equal compute also reaches $\geq 45/50$, the homotopy has no merit): plain L-BFGS with restarts reaches 0/50 at $a = 2$ (where the homotopy also has 0/50) and **50/50 at** $a = 4$, so at $a = 4$ the homotopy has no merit by the registered criterion. The restart runs at $a = 4$ succeed in 1 916 of 5 790 completed runs (33%; the budget-truncated last run of each seed is dropped; counting those 50 as misses gives 1 916/5 840, also 33%, so the survivorship bias is small), against 6/50 in C. Two unregistered checks with the same optimiser and fresh starts give 12/40 (`F1/fresh_plain_a4.py`) and 26/100 (`F1/cseeds_plain_a4.py`, C's seed formula at indices 50–149, never used by C). Fisher's exact test, two-sided, for C's 6/50: $p = 0.0012$ against the restart runs, 0.061 against 12/40, 0.058 against 26/100, 0.032 against the two fresh checks pooled (38/140). C itself reproduces: the first run of each F1b control starts from C's point and reaches the 600-cell at exactly C's six seeds (12, 15, 21, 32, 34, 42); its final energy agrees with C's to 5.5×10^{-12} (absolute; 3×10^{-14} relative) at 48 seeds, and the other two (seeds 11, 44) end in different trapped minima. The cause is the evaluator: C's own code re-run at seeds 11, 44 and 16 reproduces C's records exactly (`F1/supp_C_evaluator.py`), so the equivalent Gram-matrix form only changes floating-point paths. So C's 6/50 is a correct record that is very probably a low draw (the 26/100 check was added after seeing 12/40; only the restart comparison and the pooled one are below 0.05): the per-start success rate at $a = 4$ is nearer 0.3 than 0.12.

Control $N = 24$ (50 seeds): the homotopy ends at the 24-cell energy in 50/50 starts at every a (44/50 also match the chord at $a = 1$, where six end at the 24-cell energy to 10^{-8} with a minimum chord 1–2×10^{-6} from 1, just outside the 10^{-6} tolerance — incomplete convergence,

not a different configuration), against C's plain 25, 7, 0 of 50 and plain with restarts at equal compute 50, 47, 27 of 50; it never goes below the 24-cell. So the homotopy *does* help at $N = 24$, a configuration known not to be universally optimal, and does not at $N = 120$, where the target is proved.

Consequence (registered stop rule). F1a failed at every a; the registered rule names T1 and T2, and we extend it, as our reading, to round 1's Experiment B, which used the same kind of local search: the three-dimensional structure searches of this programme (B, T2, T1) are declared unreliable, and their negatives are read only as "no counterexample found with an unreliable search".

5.2 T2: Gaussian mixtures across the self-dual point

Kernel $e^{-a_1 r^2} + \lambda e^{-a_2 r^2}$, $(a_1, a_2) \in \{(\pi/2, 2\pi), (\pi/4, 4\pi), (2, 5)\}$; λ^* (where $E_{\mathrm{FCC}} = E_{\mathrm{BCC}}$) by bisection to 10^{-12}, then $\lambda \in \lambda^*\{0.9, 0.97, 1, 1.03, 1.1\}$; Experiment B's engine unchanged except for an appended mixture class (bit-identical on the stored references). For $a_1 a_2 = \pi^2$ the mixture with $\lambda = (\pi/a_1)^{3/2}$ is exactly self-dual ($\hat{f} = f$); the bisection recovers $\lambda^* = 2\sqrt{2}$ (to 1.9×10^{-12}) and 8 (to 2.3×10^{-10}; the limit is double-precision cancellation, since the FCC–BCC difference of the $a_2 = 4\pi$ term is only 4.8×10^{-7} against energies of order 3.5). Search: $N \in \{2, 4, 8, 12, 16\}$ with round 1's restart counts, 5 100 restarts in 75 cells. **Gate T2: 0 candidates**; no cell has anything below the best reference lattice (most negative relative difference -1.9×10^{-15}, i.e. ties), and in 14 cells (pair $(\pi/4, 4\pi)$, $N \geq 8$) the lowest energy found is above it by up to 3.7×10^{-13} relative. The FCC/BCC crossover of the mixture kernels, including the two exactly self-dual ones, is crossed without any intermediate non-lattice structure for $N \leq 16$. Hit fractions are 0.75–1.00 at λ^* (FCC and BCC are then one target) and 0.33–0.72 elsewhere, as in round 1. Remark: a fixed-density cell with $N \leq 16$ is not expected to represent the two-density coexistence that CKS list as the lowest known energy at $\alpha = \pi$ (4×10^{-6} below the pure lattices; our engine's $E_{\mathrm{FCC}}(\pi)$ is 4.1×10^{-6} above their coexistence energy, consistent), so T2 says nothing about coexistence.

5.3 T1: formally self-dual configurations at $\alpha = \pi$

Definition as implemented (radial form of [7], Sec. VI, at density 1, $Q = P$): the pair-difference measure of P (weights $1/N$) equals its structure-factor measure on Λ^* (weights $|S(w)|^2/N^2$; the pre-registration wrote $|S|^2/N$, the Poisson-consistent normalisation is N^2). Detector: the first 40 non-zero shells of each side, mismatch $< 10^{-10}$. Controls before any search: diamond 1×10^{-26} (positive), FCC↔BCC 0, $\mathbb{Z}^3$ 0, the isodual bct 9×10^{-30}, HCP 3×10^3 and FCC with itself 3.5×10^4 (negative). Search: a smooth surrogate (the identity tested on 40 Gaussians, exponents 0.94–10.5, relative residuals) minimised from 60, 60, 40 random starts for $N = 2, 4, 8$, acceptance by the registered detector. **0 accepted** (minimum mismatch 19.8, 7.7, 1.26; a polish stage minimising the shell residuals directly lowered the $N = 8$ minimum to 0.30). The grouping tolerance (10^{-6}) and weight floor (10^{-8}) of the detector are implementation choices, not registered.

Why the T1 negative is uninformative (diagnostics added after the run, unregistered). (i) The best records have surrogate $\sim 10^{-15}$ while their pair and structure-factor measures visibly differ (L1 distance of the cumulative distributions 0.7–2.4, 0 for diamond). The identity is analytic in α, so exact vanishing on an interval would force equality; what fails is numerical: 40 sampled exponents with relative residuals $\sim 10^{-8}$ admit configurations whose measures differ at $O(1)$ (the best $N = 2$ record has relative residual 1.2×10^{-10} at $\alpha = 0.3$ but 3×10^{-3} at $\alpha = 30$) — the problem is ill-conditioned. (ii) From diamond and from $\mathbb{Z}^3$ written as a 2-point cell, perturbed by 10^{-3}–0.2, the minimiser ends at surrogate $\leq 10^{-10}$ in 50/50 cases and is accepted in **0/50**. At $\varepsilon = 10^{-3}$ the *unminimised* perturbed diamond already has surrogate 3.6×10^{-11} (shell mismatch 30.5): the surrogate cannot resolve a 10^{-3} displacement from the known solution, so the minimiser has nothing to return along. (iii) The detector is exact on the controls but discontinuous: under perturbation diamond's systematically absent reflections reappear in the structure-factor measure with weights $< 5 \times 10^{-5}$ (above the 10^{-8} floor), have no counterpart in the pair measure, and shift the shell-by-shell comparison (mismatch ~ 30). The end states are nevertheless genuinely not self-dual (relative residual 1.5×10^{-5} at $\alpha = 30$, CDF distance 0.022 for the $\varepsilon = 10^{-3}$ case). T1 therefore reports a method failure, not an absence of formally self-dual configurations below FCC/BCC. A usable version needs an identifying objective or an algebraic enumeration of the CKS constructions.

5.4 H1: the two-point LP gap

Cohn–Kumar bound [6] in the periodic form $E_g(P) \geq \hat{f}(0) - f(0)$ for radial $f \leq g$ off the origin with $\hat{f} \geq 0$ (the registered transcription from [6] was not made; the statement was derived and its proof checked, `H1/RESULTS.md`, a deviation); ansatz $\hat{f}(y) = e^{-\pi|y|^2/\beta} p(|y|^2)$, $\deg p \leq 30$, in the Laguerre eigenbasis of the Fourier transform; HiGHS LP with cutting planes, 40-digit tail certificates, and an admissibility tolerance (polished violation $\leq 5 \times 10^{-9}$) fixed in code before the run — on file timestamps, not a commit — and not in the pre-registration. Sanity gate H0 in $d = 1$, where $\mathbb{Z}$ is universally optimal and the bound is sharp: at $\alpha = 1$ the bound is 0.7726372068 against $E_{\mathbb{Z}} = 0.7726372048$ ($+2 \times 10^{-9}$, inside the registered 10^{-8}, and also a measure of the discretisation error, since a valid bound cannot exceed $E_{\mathbb{Z}}$); at $\alpha = \pi$ the full-setting values are $+1.6$ to $+2.8 \times 10^{-10}$ from $E_{\mathbb{Z}}$, inside the registered 10^{-8}, but none meets the code's admissibility tolerance (violations 2–5×10^{-8}). **H0 failed under the code's admissibility tolerance (not registered); the stop then followed the registered rule**, and H1 gives no $d = 3$ value (see Sec. 6, added in version 2). The registered 10^{-8} and the code's 5×10^{-9} measure the same discretisation error from two sides (the $\alpha = 1$ value is itself 2×10^{-9} *above* $E_{\mathbb{Z}}$); part of the failure may also be a limit of the $K = 30$ ansatz rather than of certification. The H1 thresholds (10^{-6}, 10^{-3}) lie far above this 10^{-8} level, so a re-registered run with exact (rational or sum-of-squares) non-negativity certificates would probably be informative; it is left open. *Note added in version 2.* At $\alpha = \pi$ the question H1 asked was already answered in print: Cohn and de Courcy-Ireland [5] report the numerically optimised two-point bound in $d = 3$ as 0.22321782… against a record of 0.23153532…, a gap of 3.6%; we had not found this before registering H1. The follow-up H1′ (Sec. 6) reproduces it and extends it to $\alpha = 2, 4, 8$.

5.5 Round 2 deviations

F1: the run was interrupted after 6 of 50 seeds when the launching session was compacted and resumed for seeds 6–49 with the same code and single-threaded BLAS; `run_F1.py` replaces C's energy evaluator by an equivalent Gram-matrix form, asserted equal to 10^{-12} before every run (an oversubscribed first resume was stopped after 8 minutes, before any record); C's 8 000-iteration cap was used at every rung although the registration sets none ("re-converged at each rung"); the chord re-run, the Hessian check, the C-evaluator re-run, the two fresh-start checks

and the reproduction comparison with C are unregistered. By then F1a had already failed at every a (6 misses each, more than the 5 allowed). The N= 24 control completed in the first session. T2: mixture class appended; diamond and A15 added for information. T1: the N^2 normalisation; surrogate run 1 (16 Gaussians, absolute residuals) uninformative and discarded with records kept; run 2 start counts not registered (60/60/40); a typo in the bct control fixed and rerun; an earlier draft and stale code comments in `expT1.py` and `selfdual.py` misdescribed run 2's surrogate, corrected after a partial rerun reproduced 20 run-2 records bit for bit; detector tolerances not registered. H1: the admissibility tolerance was fixed in code before the run but not in the pre-registration (file timestamps only); `lp_bound.py` was touched during the run, a docstring-only change (compiled bytecode identical); tail grids and certificates added; the bound was derived rather than transcribed. All four were written up after a context compaction of the session that launched them, from the records and logs only.

6 Version 2: H1′, two-point LP values reproduced numerically

Added in version 2 (`exploration/exciton/three_d/H1p/`, pre-registration `d668109`, amendments A1 `83a3500` and A2 `968dd32`, code frozen at `91b1a97`, results `350bc74`, after review `6f5a1cb`). A literature check before re-registering found that H1's $\alpha = \pi$ question is answered in [5] (Table 1.1: two-point bound $0.08643481\ldots$, $0.15959526\ldots$, $0.22321782\ldots$ in $d = 1, 2, 3$; record $0.23153532\ldots$ in $d = 3$). H1′ was registered to reproduce these values with *rigorous* certificates (exact rational positivity of $\hat{f}$, interval arithmetic for $f \le (1+\tau)g$, bound $(\hat{f}(0)-f(0))/(1+\tau)$ by the Cohn–Kumar proposition as stated in [5] for arbitrary configurations). Development at non-gate parameters showed this does not work with our LP: the double-precision optimum is a degenerate face resolved only at its first few contacts (at $d = 1$, $\alpha = 2$ the LP controls f only up to $x \approx 3$; beyond, $(f-g)/g$ reaches 3×10^{43} at $x = 8$), an exchange step chosen in amendment A1 is ill-posed for such a face, and a high-precision Newton polish stalled. Amendment A2, committed before any gate run, replaced certification by a **numerical, non-rigorous** reproduction: H1's code unchanged, $K = 30$ and $K = 40$ (gate on $K = 40$), solutions accepted if residual violations are $\le 10^{-7}$, two-sided windows.

gate	d	α	value ($K = 40$)	reference	verdict
V1	1	1	0.7726372059	$E_{\mathbb{Z}} = 0.7726372048$ ($+1.4 \times 10^{-9}$ rel.)	pass
V1	1	π	0.0864348114	$E_{\mathbb{Z}} = 0.0864348112$ ($+2.7 \times 10^{-9}$ rel.)	pass
V2	2	π	0.1595952701	published $0.15959526\ldots$, window $[0.15959525, 0.15959528]$	pass
V3	3	π	0.2232178249	published $0.22321782\ldots$, window $[0.22321781, 0.22321784]$	pass

α	value ($K = 40$)	E_{FCC}	E_{BCC}	gap to the best lattice
2	1.011018287	1.017616896	1.015862816	4.8×10^{-3}
4	0.07275645512	0.07796767103	0.07889601371	6.7×10^{-2}
8	0.0003947635566	0.0005032413016	0.0006026980499	0.216

Reading. The values are LP objectives, not bounds: they can exceed the true optimum by far more than the residual violations (the $d = 2$ value is 3.1×10^{-9} above the triangular-lattice energy, which no valid bound can be, with a residual of 9×10^{-11}). $d = 1$ and $d = 3$ match the printed 8 digits (for $d = 3$, with the observed slack of up to $35\times$ the residual, an excess of order 2×10^{-8} cannot be excluded, comparable to the window); $d = 2$ passes only through the registered window. At $K = 30$ the $d = 3$ value would miss V3 by 4.6×10^{-7}; $K = 50$ was not run. The gaps exceed the registered 10^{-3} at all three α, as predicted: numerically, at $K \le 40$ and with the best lattice as reference, two-point bounds do not settle the three-dimensional Gaussian-core question at $\alpha = 2, \pi, 4, 8$; at $\alpha = 2$ the reading assumes no non-lattice state lies more than about 0.4% below BCC, which we did not check. Choosing β by the largest value favours the most-violating accepted solution, and at $\alpha = 8$ the β optimum is not bracketed (irrelevant at a

22% gap). This is what the literature value at $\alpha = \pi$ led us to expect, and the $\alpha = 2, 4, 8$ numbers are new only in the sense that we did not find them published. Rigorous certification at 10^{-8} remained open in version 2; version 3 supplies it with a different, interval-certified construction, after a post-hoc re-check that closed a defect in its certificate code (Sec. 7). The hostile review found no blocking item; its corrections (a residual, the "8 digits" claim for $d = 2$, the size of the slack, prior knowledge of the $K = 30$ $d = 1$ outcomes from H1, the $\alpha = 2$ reference caveat) are applied here and in `H1p/RESULTS.md`.

7 Version 3: H1″ and H1‴, certified two-point bounds

Added in version 3 (`exploration/exciton/three_d/H1pp/`, pre-registration and frozen code `b1d2de5`). Version 2 left rigorous certification open. Its obstacle was arithmetic: with the constraint rewritten as $f/g \le 1$, the $O(1)$ terms of f must cancel to the size of g (10^{-14} at the first contacts), about 23 significant digits, which double precision does not have. H1″ is the different construction that the version 2 development notes called for.

Method. Same ansatz as H1 ($\hat{f} = e^{-t/2}Q(t)$, $f = \beta^{d/2}e^{-u/2}R(u)$, Laguerre coefficients c_k, odd effective degree 39 at $K = 40$). (i) The LP is solved by a primal–dual interior-point method in 256-bit arithmetic (python-flint arb midpoints), with rows $f/g \le 1$ up to $g = 10^{-12}$, $R \le -10^{-12}$ beyond, $Q \ge 0$ on $t \in [0, \infty)$ (log grid far out). (ii) Cutting planes come from *exact* objects: after each round the coefficients are made dyadic-rational and the real roots of $R' - \kappa R$ (which contain every local maximum of f/g), of R beyond the last contact and of Q are isolated rigorously; violated regions receive rows. (iii) Every round's coefficient vector is *certified* in interval arithmetic: $R < 0$ beyond its largest real root; $f/g \le 1 + \tau_0$ on the rest by bisection with Taylor-shift enclosures; $\hat{f} + \delta e^{-\pi|y|^2/(\lambda\beta)} \ge 0$ (a Gaussian lift, $\lambda \ge 1/b$, which adds at most $\delta(\lambda\beta)^{d/2}$ to f/g), so that $f_{\mathrm{tot}} \le (1 + \tau)g$ and the Cohn–Kumar proposition in the form of [5, Prop. 2.2] gives the certified bound $B = [\hat{f}(0) + \delta - f(0) - \delta(\lambda\beta)^{d/2}]/(1 + \tau)$, whose lower interval end is reported. The best certified B over at most 25 rounds is kept (but see the defect in this certificate described below) (any certified vector is a valid bound, so the choice cannot invalidate it). Development was at non-gate parameters only and found four defects of our own before registration (a real-root filter that saw no roots, a 6×10^{-7} spike of f/g at a contact missed by a 20 000-point grid, row pruning that made violations recur, too narrow a lift); they are listed in `H1pp/DEV_NOTES.md`.

H1″ results ($K = 40$, gate on V3). V1 ($d = 1$, $\alpha = 1$ and π) certified B within 6.7×10^{-13} and 2.3×10^{-11} (relative) of $E_{\mathbb{Z}}$; V2 ($d = 2$) within 7.6×10^{-12} of E_{A_2}: pass. **V3 ($d = 3$, $\alpha = \pi$) failed**: certified $B = 0.2232177894732$, 3.05×10^{-8} below the published 0.22321782, outside the registered 10^{-8} window. By the registered stop rule the $\alpha = 2, 4, 8$ runs were reported as gated only, and the cause was left undetermined. A machine reboot killed seven running jobs after about 6.5 h; the thirteen unfinished runs were relaunched unchanged (no result was selected between launches), which is why the next round's driver checkpoints.

H1‴: larger K only (`three_d/H1ppp/`, pre-registration and frozen code `1e50f4b`). The method, row grids, certificate, b values and gate window of H1″ are unchanged; only $K = 50$ (gate) and $K = 60$ (V3 only, reported). The registered evaluator turned out to be an unmodified copy of H1″'s, which reads only $K = 40/30$ records; a replacement applying the same criteria was committed (`162dfc3`) while the runs were in flight and before any record existed, after intermediate per-round values in the logs had been seen (`H1ppp/DEVIATIONS.md`). The five runs ran uninterrupted (about 10 h at most). Certificates: $\tau \le 1.1 \times 10^{-13}$, recorded Gaussian lift $\delta \le 2 \times 10^{-28}$ ($K = 60$: $\delta = 0$ recorded, 10^{-25} after the re-check below). In three of the five runs the vectors of the last 4–5 rounds could not be certified ($\hat{f}$ check failed for every λ); the reported values come from rounds 13, 17, 13, 13 and 11, by which point the LP objective had stopped changing.

case	K	certified B	reference	verdict
V3 ($\alpha = \pi$)	40 (H1″)	0.2232177894732	published 0.22321782; -3.05×10^{-8}	fail
V3 ($\alpha = \pi$)	**50**	0.2232178193590	-6.4×10^{-10}; window edge 0.22321781	**pass (gate)**
V3 ($\alpha = \pi$)	60	0.2232178245495	$+4.5 \times 10^{-9}$	reported*
$\alpha = 2$	50	1.011018246684	$E_{\mathrm{BCC}} = 1.015862816$	$G = 4.77 \times 10^{-3}$
$\alpha = 4$	50	0.07275645543946	$E_{\mathrm{FCC}} = 0.07796767103$	$G = 6.68 \times 10^{-2}$
$\alpha = 8$	50	0.0003947602893426	$E_{\mathrm{FCC}} = 0.0005032413016$	$G = 0.216$

*Certified only after the post-hoc re-check below, with a lift $\delta = 10^{-25}$ added; B unchanged to 20 digits.

A defect in the certificate code, found by review. The hostile review of this version found that the frozen certificate (`rig2.py`/`rig3.py`) does not check $\hat{f} \geq 0$ between near-double real roots of Q (in the V3 records, pairs 2×10^{-15} to 3×10^{-14} apart, with $Q < 0$ between them) whose enclosures round to the same or adjacent doubles, and that its bisection boxes, centred at floating-point midpoints, do not cover very narrow intervals exactly. An independent re-check with exact box covering, outward-rounded root enclosures merged into clusters and the exact sign of Q in every gap (`H1ppp/recertify.py`) confirms all fourteen H1″ certificates and the four H1‴ $K = 50$ certificates with their recorded λ and lift, reproducing B to about 20 digits. The $K = 60$ record, which had no lift ($\delta = 0$), is *not* a valid certificate as recorded: Q is negative at exact rational points in 13 sub-ulp intervals (down to -9.9×10^3 near $t = 210.5$), so $\hat{f} < 0$ there. With $\delta = 10^{-25}$ it is certified and B does not change to 20 digits. "Certified" in this section therefore rests on the frozen certificate together with this re-check; the frozen code and records are unchanged.

Reading. The gate passes at $K = 50$ with a margin of 9.4×10^{-9}, so the $\alpha = 2, 4, 8$ values are registered results: *certified* lower bounds on the lower Gaussian energy of every unit-density configuration in $\mathbb{R}^3$ (energy per point summed over ordered pairs, as in [5, Def. 1.1] and in our lattice sums), 0.48%, 6.7% and 21.6% below the best lattice. Certification makes the *bound values* rigorous. It does not make rigorous the statement that two-point bounds at this degree cannot settle the three-dimensional Gaussian-core question at $\alpha = 2, 4, 8$: that would need a certified *upper* bound on the LP optimum (a dual certificate, with b optimised as well), which we did not compute. That statement remained numerical in version 3, as in version 2 (version 4 certifies it, Sec. 8) (with the same caveat at $\alpha = 2$: it assumes no non-lattice state more than about 0.4% below BCC); the registered reading of H1″, which called it certified, overstated what a lower bound can show. At $\alpha = \pi$ the reference is $\min(E_{\mathrm{FCC}}, E_{\mathrm{BCC}}) = 0.2315363$, slightly above the coexistence record 0.2315353 quoted above; this does not matter for the gate. The V3 value rises monotonically with K ($+3.0 \times 10^{-8}$ from 40 to 50, $+5.2 \times 10^{-9}$ from 50 to 60): raising K alone, with b, rows and row margin fixed, is *sufficient* to bring the certified value into the window. It does not identify the *cause* of H1″'s shortfall: H1′'s uncertified LP objective at the same $K = 40$ and b was already 0.2232178249, while H1″'s $K = 40$ objective is 0.2232177895, so H1″'s formulation (tail rows, row margin) may cost as much as the degree does, or H1′ overshot; we did not separate these. The registered prediction (gate passes, $G > 10^{-3}$ at all three, values within 10^{-6} relative of H1′'s) failed in its last part at $\alpha = 8$: the certified value is 8.3×10^{-6} below H1′'s uncertified LP objective, which most likely overshot (the H1″ certified $K = 40$ value is also about 9×10^{-6} below it). Not shown: convergence in K, the best two-point bound at $K = 50$ (b was not re-optimised), anything about three-point or semidefinite bounds.

8 Version 4: H1d, certified upper bounds on the two-point bound

Added in version 4 (`exploration/exciton/three_d/H1d/`, pre-registration and frozen code `4e596f0`, results `dd67a12`, after review `c977fa7`). Version 3's certified values are lower bounds on the two-point bound; they cannot show that the two-point bound stays below the lattice energy, which needs an *upper* bound on the LP optimum.

Method. We adapt the discrete-reduction dual of Li [11], who used it for the Cohn–Elkies sphere-packing LP, to the energy LP of [5, Prop. 2.2] at density 1, $\mathrm{LP}(\alpha) = \sup_h \hat{h}(0) - h(0)$ over continuous positive definite integrable h (so $\hat{h} \geq 0$) with $h(x) \leq g(x) = e^{-\alpha|x|^2}$ for $x \neq 0$; $\hat{G}$ denotes the discrete Fourier transform on $\mathbb{Z}_m^d$. Fix rational $s > 1$ and an integer $m \geq 2$, fold the samples of an admissible h on $s^{-1}\mathbb{Z}^d$ modulo m, $G(x) = \sum_{n \equiv x} h(n/s)$, and put $\Gamma(x) = \sum_{n \equiv x,\, n \neq 0} e^{-\alpha|n|^2/s^2}$. Poisson summation gives $\hat{G} \geq 0$ and $\hat{h}(0) \leq s^{-d} \sum_x G(x)$; $h \leq g$ off 0 gives $G \leq \Gamma$ off 0 and $-h(0) \leq -G(0) + \Gamma(0)$. For any $\lambda \geq 0$ on $\mathbb{Z}_m^d$ with $\sum \lambda = 1 - s^{-d}$ and $\mu(x) = s^{-d} + \sum_y \lambda(y) \cos(2\pi\langle x, y\rangle/m) \geq 0$ for $x \neq 0$, weak duality yields

$$\mathrm{LP}(\alpha) \;\leq\; \mathrm{UB} = \Gamma(0) + \sum_{x \neq 0} \mu(x)\Gamma(x).$$

λ comes from a floating-point LP over signed-permutation orbits (HiGHS, interior point, with $\mu \geq 10^{-8}$); it is then made exactly rational and renormalised, and μ (at every orbit) and UB are evaluated in arb with rigorous theta-sum tails. The registered class of h assumed decay $O((1 + |x|)^{-d-\delta})$ of h and $\hat{h}$; a lemma added after registration (a pre-run reviewer's argument, checked by us; Gaussian damping $h\, e^{-\pi\varepsilon|x|^2}$ and a weak form of Poisson summation) removes the decay and radial assumptions, so UB bounds the LP bound for every h admitted by [5, Prop. 2.2]. Cohn and Kumar's original, wider formulation is not claimed. A literature search (CdCI, Li, two alphaXiv searches by us; a reviewer's further alphaXiv and web searches) found dual bounds only for sphere packing; we did not find a published rigorous upper bound on the Gaussian-energy LP bound in $d = 3$, but the search was limited.

Gates and runs. V1 ($d = 1$, $\alpha = \pi$, where the LP bound is sharp): certified UB within 7.3×10^{-8} (relative) of $E_{\mathbb{Z}}$, inside the registered 10^{-6}: pass. V2 ($d = 2$, $\alpha = \pi$): consistent with the certified lower bound, 0.71% above E_{A_2} (a consistency gate only; tightness not gated). Of the 31 registered runs (3 in $d = 1$, 4 in $d = 2$, six grids (s, m) for each of the four α in $d = 3$), 29 certified; in $d = 3$ 22 of 24 certified, and two ($\alpha = \pi$ at $(8, 48)$, $\alpha = 8$ at $(6, 48)$) ended with a solver status "unknown" and gave no bound; they were not rerun. Each grid's UB is valid on its own, so reporting the smallest is not a selection effect.

α	certified lower B (v. 3)	certified upper UB	$\min(E_{\mathrm{FCC}}, E_{\mathrm{BCC}})$	UB below it by
π	0.2232178193	0.2242866642	0.2315362662	3.13%
2	1.0110182466	1.0116485641	1.0158628160	0.414%
4	0.07275645543	0.07342935691	0.07796767103	5.82%
8	0.0003947602893	0.0004109020802	0.0005032413016	18.3%

Reading. At $\alpha = 2, 4, 8$ the two-point bound in the form of [5, Prop. 2.2] lies strictly below $\min(E_{\mathrm{FCC}}, E_{\mathrm{BCC}})$, so it cannot prove FCC or BCC optimal there; this certifies the gaps that version 2 reported numerically. At $\alpha = \pi$, FCC and BCC are not the putative ground state (a coexistence state is lower [7]); there the relevant statement is that UB lies 0.00724 below the record 0.23153532 quoted in [5], so the two-point bound cannot prove that state optimal either. The interval $[B, \mathrm{UB}]$ at $\alpha = \pi$ contains the numerically optimised value printed as "0.22321782…" in [5]; the gap there was visible numerically before, and the certificate is what is new. The prediction (V1, V2 pass; all four $d = 3$ cases below $\min(E_{\mathrm{FCC}}, E_{\mathrm{BCC}})$, $\alpha = 2$ with the smallest margin; below the record at $\alpha = \pi$) held in full. A pre-run review of the mathematics and code found no blocking item; it found an evaluator flaw (a void case would not have gated the cases already evaluated), which was fixed, together with a new 10^{-12} margin on the lattice comparison, before evaluation but during the runs: 21 of the 31 records existed, one solver failure was already logged, and per-grid values had been seen (`H1d/DEVIATIONS.md`); the fix only tightens the criterion. A results review re-verified all 29 certificates with independent code in 256-bit arithmetic and found no blocking item. Not shown: anything about three-point or semidefinite bounds, the true minimal energy, other α, or where in $[B, \mathrm{UB}]$ the LP bound lies.

9 What this does and does not show

- The 3D Bravais problem for completely monotone kernels is settled numerically on our grids (FCC or BCC, exact Gaussian crossover at π); nothing new is claimed there.
- No non-Bravais structure beats the references for pure kernels up to 16 points per cell; this is an absence of counterexample with a search whose reliability is itself measured (B1 near-misses, C failures) and that round 2's stop rule, as we read it, declares unreliable.
- The "lift and project", "branes as walls" and "curved space" pictures are now each attached to a negative or marginal measurement (papers IV, this paper); the "KK tower" picture is a real and modest evaluator gain; the "T-duality" picture is a correct explanation of a negative fact.
- Round 2 adds no positive result. Its main product is a measured limit: with our optimisers, a homotopy in the kernel does not find a proved optimum on S^3 (the 600-cell) that plain restarts find at $a = 4$, the formal-duality search cannot distinguish a known formally self-dual structure from nearby non-self-dual ones, and our original LP code is not certified at the 10^{-8} level (version 2 reproduces published values numerically with it, Sec. 6; version 3 uses a new, interval-certified construction, Sec. 7). These are statements about our methods, not about the Gaussian-core conjecture.
- Version 2 (H1′) adds a reproduction, not a result: our LP code matches published two-point values numerically, and the $d = 3$ gaps at $\alpha = 2, 4, 8$ are of the size the literature led us to expect; nothing is certified.
- Version 3 (H1″, H1‴) certifies the bound values: rigorous two-point lower bounds at $K = 50$ (degree 49), validated against the published $d = 3$, $\alpha = \pi$ value after a failed attempt at $K = 40$. The gaps to the best lattice are upper estimates of the true two-point gap at this degree; that two-point bounds cannot settle these cases was not certified in version 3 (it needed a dual certificate, supplied in version 4, Sec. 8). Rigour here rests on the frozen certificate plus an independent re-check that closed a defect found in review. Nothing here proves or refutes FCC/BCC optimality.
- Version 4 (H1d) closes the direction version 3 left open: certified upper bounds show that the two-point bound (in the form of CdCI's Prop. 2.2) lies below $\min(E_{\mathrm{FCC}}, E_{\mathrm{BCC}})$ at $\alpha = 2, 4, 8$ and below the coexistence record at $\alpha = \pi$. This is a limitation of a proof method, made rigorous; it says nothing about which configuration is optimal.
- Nothing here involves the planar lattice theorem of `openai/math`, which remains AI-generated and unrefereed, nor any claim about excitons.

10 Process and deviations

Round 1 deviations: B restart counts reduced (100/100/60/40/40); A reruns after bugs with records kept; C1 operationalised as $\geq 45/50$ and the $a = 1$ rerun added after review; D's comparator defined by the run's author before running but not registered; L3's pre-registration untracked at run time; L3 annulus G2 cases run twice with the second overwriting the first; a timing check on an uncounted seed. L1: mpmath reference at 10 of 160 points; first run discarded for a reference bug. Round 2 deviations are listed with the results. All raw records, analysis scripts and the reviews (round 1, round 2, F1, in version 2 H1′, in version 3 H1‴ with its re-check script, in version 4 H1d before and after the runs) are in the repository (the H1″ review's corrections were applied in commit `25eecce`, but the review text itself was not saved); the independent review of round 1 (`docs/designs/EXCITON_FLUID_VERIFIER_NOTES_review_3d.md`) recomputed every headline number from the records and asked for the wording changes applied here (D "not met at matched cost", B1 "strictest reading", C $a = 1$ resolved by rerun).

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